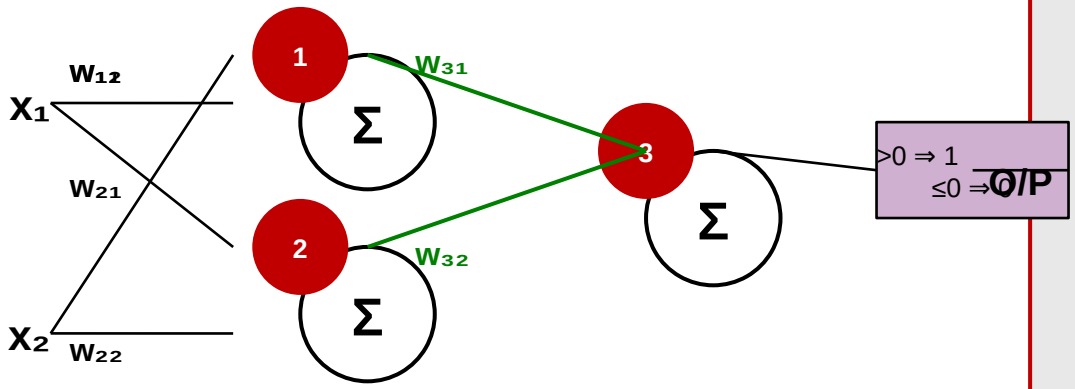
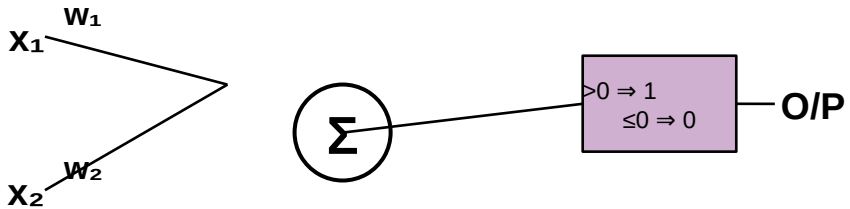
Using Two Perceptron Layers



Sum 1 = $W_{11}\, x_1 + W_{12}\, x_2$

Sum 2 = $W_{21}\, x_1 + W_{22}\, x_2$

Sum 3 = $(W_{11}\, x_1 + W_{12}\, x_2)\, w_{31} + (W_{21}\, x_1 + W_{22}\, x_2)\, w_{32}$

$= (W_{11}\, w_{31} + W_{21}\, w_{32})x_1 + (W_{12}\, w_{31} + W_{22}\, w_{32})x_2$

$= W'_1\, x_1 + W'_2\, x_2 \Rightarrow$ Still Linear Equation